RouteWise AI: Shipment Route Intelligence System

Optimizing Multi-Objective Global Freight Routing Under Uncertainty | Aiqon Take-Home Assignment

1. Executive Overview & Problem Definition

Global container shipping decisions require navigating competing, non-linear objectives under operational uncertainty. A route offering low direct freight rates frequently passes through bottleneck choke points (e.g., the Suez Canal, Bab-el-Mandeb) or congested transshipment hubs (Singapore, Rotterdam), multiplying delay variance. Conversely, express multimodal routing provides high speed and schedule reliability at prohibitive cost premiums. **RouteWise AI** formulates shipment routing as a constrained multi-criteria decision analysis (MCDA) optimization problem paired with an explainable AI layer. Rather than treating route selection as an opaque black box, the system computes transparent, reproducible trade-off vectors and grounds an LLM assistant strictly in deterministic optimization outcomes.

2. Mathematical Formulation & Decision Architecture

Candidate routes are evaluated across three normalized core dimensions: Landed Cost (C), Transit Time (T), and Composite Risk (R). To ensure fair comparison across diverse trade corridors, attributes are normalized dynamically relative to each shipment's candidate pool.

Normalization & Composite Scoring:

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C_norm = (Cost - Cost_min) / max(Cost_max - Cost_min, ε)
T_norm = (Time - Time_min) / max(Time_max - Time_min, ε)
R_comp = 0.40 * P(delay) + 0.30 * (Congestion/100) + 0.30 * (Geopolitical/100)
Score_total = w_c * C_norm + w_t * T_norm + w_r * R_norm (where w_c + w_t + w_r = 1.0)
```

Lower score indicates superior multi-objective performance. Unlike naive rank-averaging, this continuous formulation maintains proportionality: an option that is only 2% more expensive but 40% faster receives an appropriately weighted advantage.

3. System Architecture: Decoupled Optimization vs. Generative AI

A fundamental design principle of RouteWise AI is the **strict separation of mathematical decision-making from natural language generation**. Large Language Models (LLMs) are notorious for stochastic arithmetic errors, boundary hallucinations, and inconsistent numerical ranking. Consequently, the system operates in two decoupled tiers:

Component Layer	Technology / Implementation	Operational Responsibility
Deterministic Optimizer	Python, NumPy, pandas	Hard constraint enforcement, Pareto dominance calculation, normalization, continuous score synthesis.
Explainability Engine	Rule-Based Engine + Gemini LLM	Synthesizes audit-ready natural-language rationale, explains why runner-up lost, answers interactive what-if queries.
Interactive Dashboard	Streamlit, Plotly	Interactive parameter tuning, trade-off radar / scatter visualizations, scenario sensitivity analysis.

4. Constraint Handling, Pareto Dominance & Robustness

Real-world logistics systems encounter tight customer service level agreements (SLAs) and strict budget caps. RouteWise AI addresses these through deterministic mechanisms:

1. Hard Constraint Invalidation: Routes violating max budget (C_i > C_max), max transit days (T_i > T_max), or max risk thresholds are immediately quarantined with explicit diagnostic failure messages (e.g., *'Exceeds max transit time by 4.5 days'*). They are disqualified from winning unless *all* candidates are infeasible.
2. Graceful Infeasible Recovery: If an operator specifies an impossible constraint set, the engine flags all_infeasible = True and returns a transparent 'best-effort' recommendation ranked by minimal penalty distance rather than throwing an unhandled exception.
3. Pareto Dominance Filtering: Route A strictly dominates Route B (A ■ B) if C_A ≤ C_B, T_A ≤ T_B, and R_A ≤ R_B with at least one strict inequality. Dominated routes are flagged for logistics planners, providing mathematically verifiable justification for early route pruning.
4. Zero-Division & Identical Metric Safeguard: An ε = 10■■ threshold guarantees numerical stability when all route options in a corridor share identical attributes.

5. Empirical Evaluation & Sensitivity Benchmark

To test system behavior across diverse global freight dynamics, a synthetic benchmark dataset of 30 shipments across major corridors (Trans-Pacific, Asia-Europe, South Asia-Europe, Intra-Med) with 4 route archetypes each (120 options total) was generated. Each corridor includes: (1) Economic Standard Ocean, (2) Express Direct Ocean, (3) Resilient Cape Diversion, and (4) Multimodal Sea-Air Express.

Scenario / Profile	Weight Profile (C/T/R)	Dominant Winner Archetype	Observed Operational Dynamics
Balanced Baseline	40% / 30% / 30%	Express Direct Ocean	Balances moderate cost premium with 25% transit time reduction and low congestion.
Cost-Focused	70% / 15% / 15%	Economic Standard Ocean	Captures lowest freight rate; accepts 7-10 extra days transit and higher port delay risk.
Time-Critical	15% / 70% / 15%	Multimodal Sea-Air Express	Drastically cuts transit from 28d to 8d; incurs 2.2x cost acceptable for critical JIT components.
Risk-Averse	15% / 15% / 70%	Resilient Cape Diversion	Bypasses geopolitical conflict zones (Red Sea); lower delay probability despite longer nautical path.

Key Sensitivity Finding: *Shifting from a Balanced to a Risk-Averse profile dynamically flips the optimal routing from Suez/Red Sea transshipment to the Cape of Good Hope corridor without human intervention, proving the engine's sensitivity to real-world maritime disruption scenarios.*

6. Explainability Framework & Grounded AI Verification

A critical evaluation metric in logistics decision support is auditability: operators will not trust recommendations they cannot explain to customers. RouteWise AI implements a dual-mode explainability architecture:

• **Grounded Deterministic Engine (Zero-Shot Fallback):** Translates exact delta vectors into executive explanations without network or API dependencies. It explicitly reports percentage cost differences, transit delta days, risk factors, and reasons why alternatives were rejected.
• **Grounded LLM Prompting:** When an API key is present, the verified mathematical result (JSON payload containing scores, trade-off deltas, and constraints) is injected into an LLM prompt instructed to act as an executive logistics advisor. Hallucination is strictly bounded because the LLM is prohibited from inventing values.

7. Practical System Limitations & Production Roadmap

While RouteWise AI provides robust multi-objective decision support, real-world deployment requires addressing key logistics realities:

Limitation	Root Cause in Practice	Production Mitigation Strategy
Static Risk Indices	Synthetic congestion and geopolitical scores represent point-in-time snapshots.	Integrate live AIS satellite telemetry and port dwell-time APIs (e.g., MarineTraffic, Spire).
Non-Linear Demurrage	Delay costs are rarely linear; exceeding free-time windows causes step-function detention spikes.	Implement piecewise linear demurrage penalty functions conditioned on bill-of-lading terms.
Deterministic vs. Stochastic	Assumes fixed transit time estimates despite weather and labor strike volatility.	Upgrade scoring engine with Monte Carlo simulation and stochastic chance-constrained programming.
Carbon / Scope 3 ESG	Emissions currently treated as an informative attribute rather than an optimization penalty.	Add carbon cap constraints (IMO CII regulations) directly into the multi-objective Pareto solver.

8. Conclusion & Deliverables Summary

RouteWise AI demonstrates that combining deterministic optimization algorithms with grounded LLM explanation layers yields a reliable, practical, and explainable decision-support tool for global logistics planners. All deliverables—including clean source code, 100% passing test suite, interactive Streamlit dashboard, and reproducible synthetic dataset—are packaged for evaluation in the GitHub repository.